WARPSimLab Historical Windows Risk Report

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Projection Period: 2026-2056 (30 Years)

Report Basis: Real Dollars (Inflation Adjusted)

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Executive Summary

This section summarizes the main risk results from the selected analysis method.

Scenario Count

121

Years Simulated

30

Simulated Shortfall Rate

69.42%

Worst Ending Portfolio

$0

Median Ending Portfolio

$0

Best Ending Portfolio

$929,801

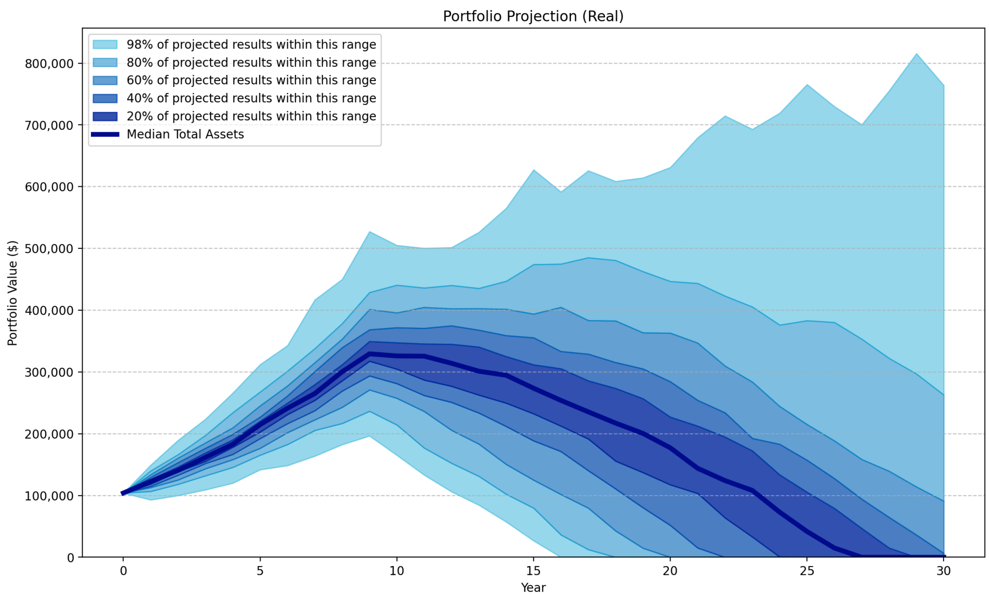
Method Explanation

Historical Window Analysis evaluates portfolio outcomes using rolling periods of actual historical market and inflation data. Each path represents one historical economic sequence applied over the modeled period.

This method is useful for understanding how different historical market environments would have affected your financial plan.

Portfolio Projection Across Historical Market Periods

This chart summarizes portfolio outcomes produced by applying actual historical market returns and inflation to your financial plan. Each scenario begins in a different historical year, allowing the report to compare how the modeled portfolio would have behaved under real market conditions experienced in the past.



Portfolio Sustainability Analysis

This section summarizes how often the modeled portfolio was depleted before the end of the projection period.

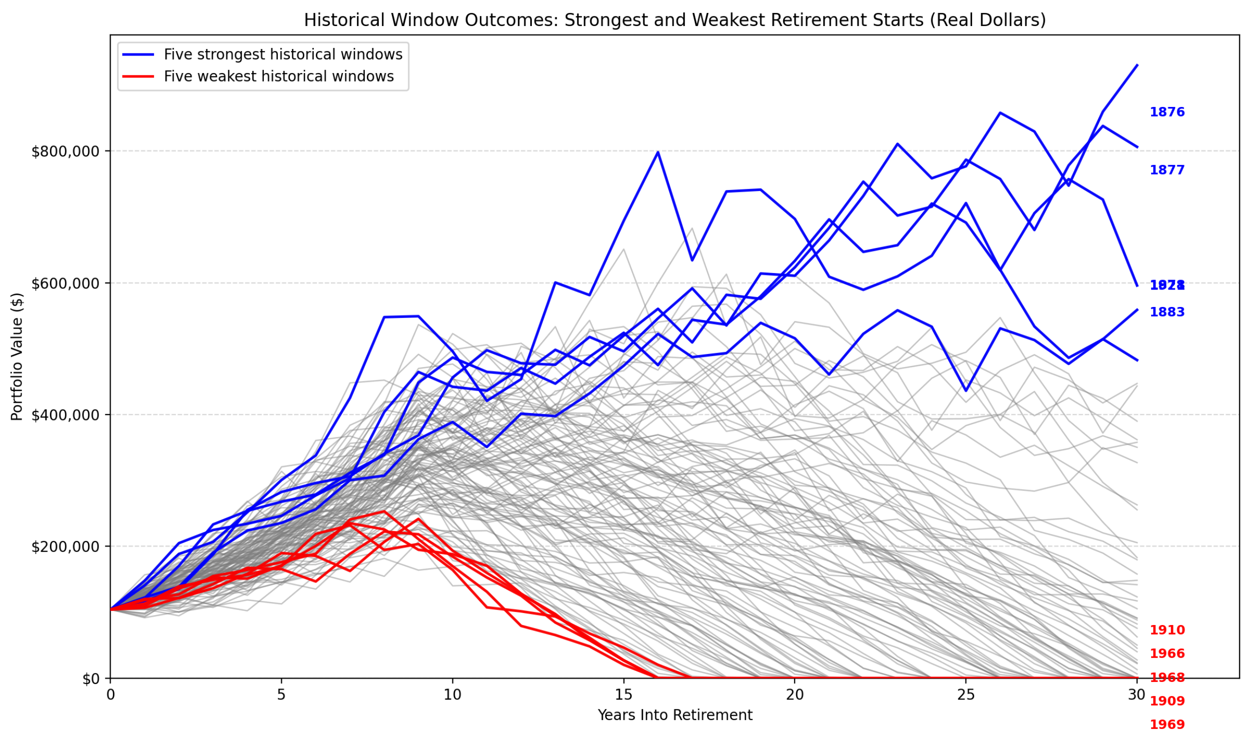
Historical Window Analysis evaluates your financial plan using actual historical sequences of market returns and inflation. Because every scenario represents conditions that actually occurred, the results illustrate how your plan would have performed across a wide range of historical market environments.

Scenario Count	121
Portfolio Survived Count	37
Portfolio Depleted Count	84
Portfolio Survival Rate	30.58%
Simulated Shortfall Rate	**69.42%**
Earliest Portfolio Depletion Year	**2,042**
Median Portfolio Depletion Year	**2,049**
Latest Portfolio Depletion Year	**2,056**

Historical Window Insights

Historical Window Analysis evaluates your financial plan by analyzing numerous scenarios using rolling periods of actual historical market returns and inflation data. Each scenario applies one historical sequence of market returns and inflation to your financial plan.

This method illustrates how your financial plan would have performed across a wide range of real-world market environments, including periods of strong market growth, prolonged downturns, elevated inflation, and challenging early-return sequences.



The chart shows all historical portfolio paths in light gray. The five strongest historical retirement starts are redrawn in blue, while the five weakest are redrawn in red. The labels at the right edge identify the historical retirement start year for each highlighted path.

Sequence-of-Returns Risk

Historical Window Analysis naturally incorporates sequence-of-returns risk because each historical scenario follows the actual chronological order of historical market returns. Early bear markets generally place greater stress on portfolios than identical losses occurring later in retirement.

Best Historical Retirement Start Years

Worst Historical Retirement Start Years

These retirement start years produced the strongest long-term portfolio outcomes.

- **1876** - Portfolio Sustained, $929,801
- **1877** - Portfolio Sustained, $806,089
- **1878** - Portfolio Sustained, $595,751
- **1921** - Portfolio Sustained, $558,756
- **1883** - Portfolio Sustained, $482,515

These retirement start years produced the weakest long-term portfolio outcomes.

- **1969** - Portfolio Depleted, $0
- **1909** - Portfolio Depleted, $0
- **1968** - Portfolio Depleted, $0
- **1966** - Portfolio Depleted, $0
- **1910** - Portfolio Depleted, $0

What History Suggests

Historical market conditions produced significant portfolio sustainability risk. More than half of the evaluated historical sequences depleted the portfolio before the end of the projection period.

The weakest historical paths reached portfolio depletion. These scenarios illustrate sequence-of-returns risk: poor investment returns early in retirement can have lasting effects because withdrawals continue while the portfolio is under pressure.

Unlike Monte Carlo Analysis, Historical Window Analysis evaluates your financial plan using actual historical market and inflation sequences. This makes it useful for understanding how specific historical market environments influenced your projected financial outcomes.

Percentile Portfolio Table

This table summarizes how portfolio values varied across all simulated scenarios during each year of the simulation. Each row represents one year of the projection, while the percentile columns show the distribution of portfolio values across all historical windows or Monte Carlo simulations at that point in time.

For example, the 10th percentile represents a relatively poor outcome in which approximately 10% of scenarios produced a lower portfolio value and 90% produced a higher value. Likewise, the median represents the middle outcome, while the 90th percentile represents one of the stronger outcomes. Together, these percentiles illustrate how the range of possible portfolio values changes over time.

Year	10th Percentile	25th Percentile	Median	75th Percentile	90th Percentile
2026	$104,000	$104,000	$104,000	$104,000	$104,000
2027	$106,719	$113,993	$122,749	$130,885	$140,066
2028	$118,198	$128,552	$140,654	$154,990	$166,340
2029	$132,037	$149,013	$161,663	$179,807	$197,573
2030	$145,638	$162,115	$182,958	$205,027	$234,275
2031	$165,379	$184,363	$214,767	$235,068	$267,712
2032	$182,622	$208,280	$241,244	$265,248	$301,559
2033	$205,467	$233,507	$265,019	$308,436	$337,988
2034	$216,750	$256,655	$300,276	$341,694	$378,394
2035	$236,506	$288,985	$329,327	$382,049	$428,929
2036	$214,305	$270,443	$325,945	$385,109	$440,721
2037	$176,961	$253,583	$325,539	$384,740	$436,135
2038	$152,283	$231,573	$314,066	$389,770	$440,285
2039	$131,125	$205,158	$301,200	$395,446	$435,436
2040	$102,135	$190,976	$294,637	$378,253	$447,134
2041	$79,683	$160,243	$273,662	$367,874	$474,103
2042	$36,319	$119,099	$253,969	$355,127	$474,892
2043	$12,485	$106,106	$235,400	$353,589	$485,155
2044	$0	$80,151	$217,201	$332,261	$480,699

Year	10th Percentile	25th Percentile	Median	75th Percentile	90th Percentile
2045	$0	$43,331	$200,496	$329,345	$462,691
2046	$0	$11,124	$177,109	$314,523	$446,717
2047	$0	$0	$144,014	$287,500	$443,574
2048	$0	$0	$124,093	$264,119	$423,038
2049	$0	$0	$108,190	$250,423	$405,411
2050	$0	$0	$72,960	$213,918	$376,146
2051	$0	$0	$41,677	$193,252	$383,228
2052	$0	$0	$14,996	$157,861	$380,361
2053	$0	$0	$0	$130,197	$353,318
2054	$0	$0	$0	$112,884	$322,235
2055	$0	$0	$0	$86,558	$296,936
2056	$0	$0	$0	$45,250	$262,836

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